

Gaurav Prasad

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Early-career software engineer focused on high-performance C++ systems, simulation, and realtime data processing. Built multithreaded graphics, physics, and market-data systems with an emphasis on concurrency, performance optimization, and interactive visualization.

EDUCATION

University of Colorado Boulder

Graduating June 2026

B.S. in Information Science | Minor in Computer Science

Boulder, CO

PROJECTS

Parallel MPM N-Body Fluid Simulator | [Demo](#)

May 2026 - Current

- Built a real-time 2D Material Point Method simulator in C++20 and OpenGL with particle-grid transfers, APIC/FLIP velocity updates, pressure, viscosity, and deformation tracking.
- Parallelized particle-grid passes with cached weight stencils and tiled simulation passes using C++ execution policies and per-tile accumulators, supporting interactive runs with up to 500,000 particles at 60fps.
- Implemented dual Gravity + Radiance Cascades to efficiently simulate far field self-gravity and global illumination for dense, temperature-driven clusters of diverse materials.
- Added GLSL particle and radiance rendering toggles, frame-phase timing instrumentation, and interactive spawning controls for testing and debugging diverse material and light behavior.

Real-Time Local Order Book Engine | [Github](#)

April 2026 - May 2026

- Built a multithreaded C++17 market data system using Boost.Asio/Beast websockets to maintain a synchronized local Binance order book in real time with incremental update processing times averaging 8-20 μ s.
- Implemented REST snapshot + incremental depth-update synchronization logic to ensure deterministic recovery and consistent book state under streamed updates.
- Designed concurrent ingestion, processing, and rendering pipelines using mutex-protected shared snapshots and asynchronous networking.
- Computed and visualized orderbook alongside market microstructure metrics including spread, midpoint price, and microprice from live order flow.

SPH Wind Tunnel Simulation | [Demo](#)

February 2026 - April 2026

- Developed a real-time smoothed particle hydrodynamics wind tunnel simulator in C++17 to visualize dense 2D fluid flow around obstacles with velocity-based color mapping.
- Accelerated local particle interactions with a uniform-grid spatial partition using cache friendly localized particle storage and 3x3 local neighborhood queries for 200k+ particles.
- Parallelized density, force accumulation, and integration passes with C++ execution policies reducing frame time from ~150ms to ~16ms at 200k particles.
- Leveraged ImGui to monitor frametimes and create runtime controls for adjusting simulation variables including viscosity and vorticity to simulate different fluid conditions.

EXPERIENCE

FAS 360 Summer Intern GSA Professional Services and Human Capital (GSA.gov)

Summer 2024

- Supported HR and program operations for a 12-week federal internship program, coordinating training logistics, scheduling, and cross-team communication.
- Assisted with internal training delivery, meeting coordination, and documentation workflows in a structured public-sector environment.

Visualization Consultant Inclusive America

January 2024 - May 2024

- Developed interactive Power BI dashboards to visualize diversity metrics across U.S. state governments for leadership and external stakeholder review.
- Created repeatable documentation and handoff materials that enabled the workflow to scale from pilot states to the remaining 48 states with minimal rework.

TECHNICAL EXPERIENCE

Languages

C++, C, GLSL, Python, SQL

Systems & Performance

Multithreading, Concurrency, Real-Time Systems, Numerical Simulation, Spatial Partitioning, Performance Optimization, Networking

Tools & Libraries

Linux, Git, CMake, Make, OpenGL, ImGui, Boost.Asio/Beast, OpenCV